

Semester 1 (2025-2026)
AI306DE01 - Data Analytics
FINAL PROJECT

- i. **Due date:** 24/12/2024
- ii. **Submission:** on *mlearning*.
 - Only 1 member of the group should submit group's work, but **ALL** of member's names must be included in the report (Fill in group members in the spreadsheet on mlearning)
 - List each member's contribution on a scale **0% - 100%**
 - No hardcopies required.
- iii. **Format:** must include files:
 - **FullName_report.docx** and **FullName_report.pdf** should include:
 - a. Format: HSU templates (if applicable)
 - b. Length: 15+ pages
 - c. Theory of **cleaning** data, handling **missing data**, **data scaling**, **normalization**
 - d. Theory of **One-hot encoding**
 - e. Theory of **Outlier detection**
 - f. Theory of **Feature Importance/ Feature Ranking**
 - g. Plots/Charts of the data
 - h. **Analytical results and conclusion**
 - **slides.pptx**
 - **source_code.ipynb**
 - all used **.csv** files
- iv. Group of 3-5 students
- v. **Dataset:** You can choose one of the 5 datasets below:
 -  bank_loan.csv
 -  credit_card.csv
 -  gender_voice.csv
 -  healthy.csv
 -  heart_disease.csv

vi. **Requirements:** Data Analytics and Analysis

The report must include all of the following (at least 10+/12 criteria):

- 1 Must include **cleaning** data
- 2 Must include handling **missing data (MICE algorithm)**
- 3 (Must) include **data scaling/normalization**
- 4 (Must) include **one-hot encoding**
- 5 Must include **Feature Importance/ Feature Ranking**
- 6 Must include **outlier detection**
- 7 Must include **correlation matrices and heat maps**
- 8 Must include a Regression/or Classification model
- 9 Must include Accuracy, Precision, Recall, F1 Score, ... /Confusion matrix
- 10 (Must) include a comparison
- 11 (Must) include **ROC** (Receiver Operating Characteristic) curve
- 12 Must include **Visualization (charts/plots)** of at least 5 types (scatter, line, bar, pie chart, boxplot, histogram, ...)

--- The End---